

Data Analytics Project 2 – Exploratory Data Analysis (EDA) on E-Commerce Dataset

Submitted By:

Yanamala Manisha

Internship:

DecodeLabs Data Analytics Internship

Date:

June 2026

Project Objective:

- Perform Exploratory Data Analysis (EDA) on an E-Commerce dataset. Understand patterns, trends, and distributions.
- Generate meaningful insights using SQL queries.

Tools Used:

- MySQL Workbench

Dataset Information:

- **Dataset:** E-Commerce Sales Dataset
- **Records:** 1000
- **Data Type:** Structured Data

Main Columns

- OrderID
- CustomerID
- Product
- Quantity
- UnitPrice
- TotalPrice
- PaymentMethod
- OrderStatus
- ReferralSource

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: project project SQL File 4* SQL File 5* x

SCHEMAS

Filter objects

▼ **deodelabs_project1**

- Tables
- Views
- Stored Procedures
- Functions

► parks_and_recreation

► sys

Administration Schemas

Information

No object selected

Object Info Session

```
1 • use deodelabs_project1;
2 Open a script file in this editor
3 -- VIEW DATASET
4 • SELECT *
5 FROM ecommerce_data;
```

Result Grid

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
►	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	No Coupon Applied	Instagram	2853.1
	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	No Coupon Applied	Referral	302.7
	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	No Coupon Applied	Email	2753.4
	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	No Coupon Applied	Facebook	273.19
	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	No Coupon Applied	Email	2504.04
	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	No Coupon Applied	Instagram	491.72
	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	No Coupon Applied	Facebook	664.42
	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	No Coupon Applied	Facebook	747.75
	ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	No Coupon Applied	Email	268.56
	ORD200009	2023-11-21	C31946	Desk	4	500.38	102 Main St	Credit Card	Shipped	TRK73478363	6	No Coupon Applied	Google	2037.52

ecommerce_data 1 x

Output

Action Output

#	Time	Action	Message	Duration
✓ 1	12:57:33	use deodelabs_project1	0 row(s) affected	0.000 s
✓ 2	12:58:18	SELECT * FROM ecommerce_data LIMIT 0, 1000	1000 row(s) returned	0.016 s

Hot weather Now

Search

ENG IN

Fig 1: Dataset

Statistics:

Analysis Performed:

- Total Records
- Average Order Value
- Highest Order Value
- Lowest Order Value

Results:

- **Highest Order Value: 3456.40**

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: project project SQL File 4* SQL File 5* x

SCHEMAS

Filter objects

▼ decode labs_project1

- Tables
- Views
- Stored Procedures
- Functions
- parks_and_recreation
- sys

Administration Schemas

Information

Schema: decode labs_project1

```
13 FROM ecommerce_data;
14
15
16 -- MAXIMUM ORDER values
17 • SELECT MAX(TotalPrice) AS Highest_Order_Value
18 FROM ecommerce_data;
19
20
21
22
```

Result Grid

Highest_Order_Value
3456.4

Result 11 x

Output

Action Output

#	Time	Action	Message	Duration /
✓ 4	13:07:33	SELECT AVG(TotalPrice) AS Average_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.015 sec.
✓ 5	13:10:04	SELECT MAX(TotalPrice) AS Highest_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.016 sec.
✓ 6	13:15:09	SELECT MIN(TotalPrice) AS LOWEST_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.000 sec.
✓ 7	13:40:59	SELECT COUNT(*) AS TOTAL_RECORDS FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.016 sec.
✓ 8	13:42:56	SELECT AVG(TotalPrice) AS Average_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.016 sec.
✓ 9	13:44:23	SELECT MAX(TotalPrice) AS Highest_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.000 sec.

Object Info Session

35°C Partly sunny

Search

ENG IN

Fig 2 : Maximum Order Value

- **Lowest Order Value: 11.39**

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: project project SQL File 4* SQL File 5*

SCHEMAS

Filter objects

▼ deodelabs_project1

Tables

Views

Stored Procedures

Functions

► parks_and_recreation

► sys

Administration Schemas

Information

Schema: deodelabs_project1

Result Grid

Filter Rows: Export: Wrap Cell Content:

LOWEST_Order_Value

► 11.39

Result 12 x

Output

Action Output

#	Time	Action	Message	Duration /
5	13:10:04	SELECT MAX(TotalPrice) AS Highest_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.016 sec.
6	13:15:09	SELECT MIN(TotalPrice) AS LOWEST_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.000 sec.
7	13:40:59	SELECT COUNT(*) AS TOTAL_RECORDS FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.016 sec.
8	13:42:56	SELECT AVG(TotalPrice) AS Average_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.016 sec.
9	13:44:23	SELECT MAX(TotalPrice) AS Highest_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.000 sec.
10	13:45:42	SELECT MIN(TotalPrice) AS LOWEST_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.015 sec.

Object Info Session

35°C Partly sunny

Search

ENG IN

Fig 3 : Minimum Order Value

Product Analysis:

Queries Performed:

- Top Selling Products
- Revenue by Product

Observation:

- Instagram generated the highest customer traffic.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: project project SQL File 4* SQL File 5*

SCHEMAS

Filter objects

decode labs_project1

Tables Views Stored Procedures Functions parks_and_recreation sys

Administration Schemas

Information

Schema: decode labs_project1

Result Grid

Filter Rows: Export: Wrap Cell Content:

PRODUCT	Total_Sold
Chair	562
Printer	542
Laptop	535
Desk	508
Tablet	497
Monitor	480
Phone	411

Result 13 x

Output

Action Output

#	Time	Action	Message	Duration /
6	13:15:09	SELECT MIN(TotalPrice) AS LOWEST_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.000 sec.
7	13:40:59	SELECT COUNT(*) AS TOTAL_RECORDS FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.016 sec.
8	13:42:56	SELECT AVG(TotalPrice) AS Average_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.016 sec.
9	13:44:23	SELECT MAX(TotalPrice) AS Highest_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.000 sec.
10	13:45:42	SELECT MIN(TotalPrice) AS LOWEST_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.015 sec.
11	13:50:02	SELECT PRODUCT, SUM(QUANTITY) AS Total_Sold FROM ecommerce_data GROUP BY Product ORDER...	7 row(s) returned	0.015 sec.

Object Info Session

35°C Partly sunny

Search

ENG IN

Fig 4 : Top Selling Products

Payment Analysis:

- Orders by Payment Method
- Online
- Cash
- Credit Card
- Debit Card
- Gift Card
- Observation
- Online payments recorded the highest number of orders.

Order Status Analysis:

- Order Status
- Shipped
- Delivered
- Cancelled
- Returned
- Pending
- Observation
- Cancelled orders were slightly higher than the other order statuses.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: project project SQL File 4* SQL File 5*

Limit to 1000 rows

SCHEMAS

Filter objects

▼ decode labs_project1

- Tables
- Views
- Stored Procedures
- Functions
- ▼ parks_and_recreation
- ▼ sys

Administration Schemas

Information

Schema: decode labs_project1

```
42
43 -- ORDER STATUS ANALYSIS
44 • SELECT OrderStatus,
45    COUNT(*) AS Total_Orders
46 FROM ecommerce_data
47 GROUP BY OrderStatus;
48
49
50
51
```

Result Grid

OrderStatus	Total_Orders
Shipped	235
Cancelled	250
Returned	247
Delivered	231
Pending	237

Result 16 x

Output

Action Output

#	Time	Action	Message	Duration /
✓ 9	13:44:23	SELECT MAX(TotalPrice) AS Highest_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.000 sec.
✓ 10	13:45:42	SELECT MIN(TotalPrice) AS LOWEST_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.015 sec.
✓ 11	13:50:02	SELECT PRODUCT, SUM(QUANTITY) AS Total_Sold FROM ecommerce_data GROUP BY Product ORDER...	7 row(s) returned	0.015 sec.
✓ 12	13:55:01	SELECT PRODUCT, SUM(TotalPrice) AS Revenue FROM ecommerce_data GROUP BY Product ORDER BY...	7 row(s) returned	0.015 sec.
✓ 13	13:59:31	SELECT PaymentMethod, COUNT(*) AS Total_Orders FROM ecommerce_data GROUP BY PaymentMethod ...	5 row(s) returned	0.015 sec.
✓ 14	14:04:16	SELECT OrderStatus, COUNT(*) AS Total_Orders FROM ecommerce_data GROUP BY OrderStatus LIMIT 0, ...	5 row(s) returned	0.000 sec.

Object Info Session

36°C Partly sunny

Search

ENG IN

Fig 5: Payment Analysis

Referral Source Analysis:

- Referral Sources
- Instagram
- Email
- Google
- Facebook
- Referral
- Observation
- Instagram generated the highest customer traffic.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

▼ deodelabs_project1

- Tables
- Views
- Stored Procedures
- Functions
- parks_and_recreation
- sys

Administration Schemas

Information

Schema: deodelabs_project1

Object Info Session

project project SQL File 4* SQL File 5* x

Limit to 1000 rows

```

48
49 -- REFERRAL SOURCE ANALYSIS
50 • SELECT ReferralSource,
51 COUNT(*) AS Total_Customers
52 FROM ecommerce_data
53 GROUP BY ReferralSource
54 ORDER BY Total_Customers DESC;
55
56
57

```

Result Grid

ReferralSource	Total_Customers
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Result 17 x

Output

Action Output

#	Time	Action	Message	Duration /
✓ 10	13:45:42	SELECT MIN(TotalPrice) AS LOWEST_Order_Value FROM ecommerce_data LIMIT 0, 1000	1 row(s) returned	0.015 sec
✓ 11	13:50:02	SELECT PRODUCT, SUM(QUANTITY) AS Total_Sold FROM ecommerce_data GROUP BY Product ORDER...	7 row(s) returned	0.015 sec
✓ 12	13:55:01	SELECT PRODUCT, SUM(TotalPrice) AS Revenue FROM ecommerce_data GROUP BY Product ORDER BY...	7 row(s) returned	0.015 sec
✓ 13	13:59:31	SELECT PaymentMethod, COUNT(*) AS Total_Orders FROM ecommerce_data GROUP BY PaymentMethod ...	5 row(s) returned	0.015 sec
✓ 14	14:04:16	SELECT OrderStatus, COUNT(*) AS Total_Orders FROM ecommerce_data GROUP BY OrderStatus LIMIT 0, ...	5 row(s) returned	0.000 sec
✓ 15	14:10:45	SELECT ReferralSource, COUNT(*) AS Total_Customers FROM ecommerce_data GROUP BY ReferralSource...	5 row(s) returned	0.015 sec

1 cm of rain Wednesday

Search

ENG IN

Fig 6 : Referral Source Analysis

Monthly Sales Trend:

- Sales Trend Analysis
- Monthly Revenue
- Sales Distribution
- Observation
- Sales varied across different months.

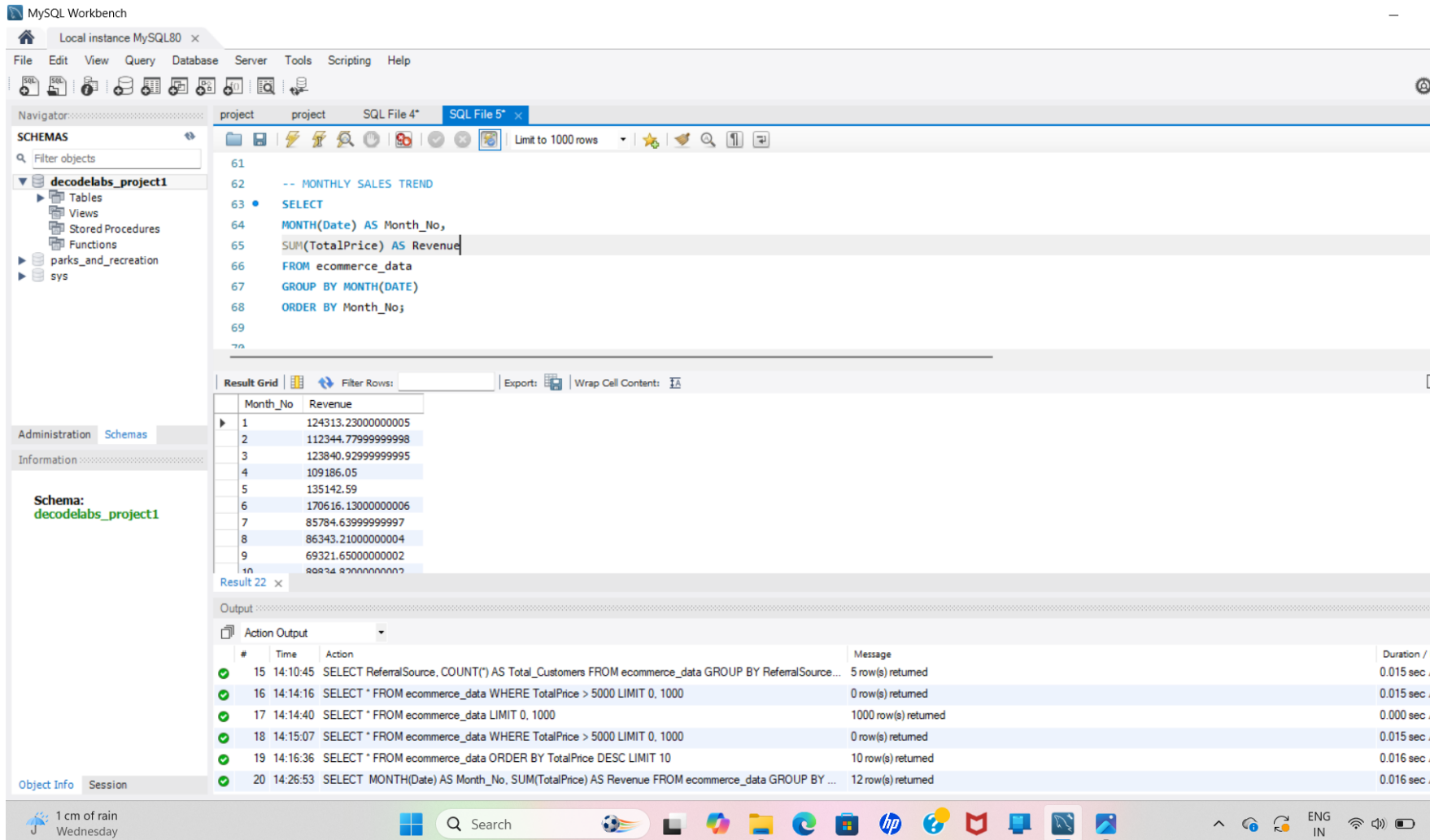


Fig 7 :Monthly Sales Trend

Outlier Detection:

- Outlier Analysis
- Checked for unusually high-value orders.
- Observation
- No significant outliers above the selected threshold were found.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: project project SQL File 4* SQL File 5* x

Limit to 1000 rows

SCHEMAS

Filter objects

▼ deodelabs_project1

- Tables
- Views
- Stored Procedures
- Functions
- parks_and_recreation
- sys

Administration Schemas

Information

Schema: deodelabs_project1

```
55
56 -- OUTLIER DETECTION (HIGH VALUE ORDERS)
57 • SELECT *
58 FROM ecommerce_data
59 ORDER BY TotalPrice DESC
60 LIMIT 10;
61
62
63
```

Result Grid

Filter Rows: Export: Wrap Cell Content: Fetch rows:

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200789	2023-08-17	C57276	Tablet	5	691.28	183 Main St	Online	Delivered	TRK75899752	10	No Coupon Applied	Email	3456.4
ORD201122	2023-06-07	C38840	Monitor	5	678.19	766 Main St	Online	Returned	TRK32496970	8	No Coupon Applied	Facebook	3390.95
ORD200632	2023-05-02	C67260	Laptop	5	678.16	463 Main St	Gift Card	Delivered	TRK38229104	7	No Coupon Applied	Facebook	3390.8
ORD200469	2023-11-26	C13877	Chair	5	676.98	893 Main St	Cash	Cancelled	TRK17254691	5	No Coupon Applied	Facebook	3384.9
ORD200328	2023-02-28	C18404	Tablet	5	674.04	546 Main St	Online	Cancelled	TRK89401624	7	No Coupon Applied	Google	3370.2
ORD200107	2023-03-27	C16775	Printer	5	670.75	848 Main St	Gift Card	Shipped	TRK34392124	8	No Coupon Applied	Instagram	3353.75
ORD200326	2024-07-01	C65986	Laptop	5	670.48	273 Main St	Gift Card	Returned	TRK98353867	5	No Coupon Applied	Facebook	3352.4
ORD201065	2023-10-30	C47778	Printer	5	666.8	488 Main St	Debit Card	Delivered	TRK79504329	7	No Coupon Applied	Referral	3334
ORD201031	2023-02-28	C59183	Phone	5	664.51	136 Main St	Debit Card	Pending	TRK18129706	8	No Coupon Applied	Email	3322.55
ORD200463	2023-05-26	C25276	Laptop	5	662.78	214 Main St	Debit Card	Shipped	TRK14011732	9	No Coupon Applied	Instagram	3312.9

ecommerce_data 21 x

Output

Action Output

#	Time	Action	Message	Duration /
✓ 14	14:04:16	SELECT OrderStatus, COUNT(*) AS Total_Orders FROM ecommerce_data GROUP BY OrderStatus LIMIT 0, ...	5 row(s) returned	0.000 sec.
✓ 15	14:10:45	SELECT ReferralSource, COUNT(*) AS Total_Customers FROM ecommerce_data GROUP BY ReferralSource...	5 row(s) returned	0.015 sec.
✓ 16	14:14:16	SELECT * FROM ecommerce_data WHERE TotalPrice > 5000 LIMIT 0, 1000	0 row(s) returned	0.015 sec.
✓ 17	14:14:40	SELECT * FROM ecommerce_data LIMIT 0, 1000	1000 row(s) returned	0.000 sec.
✓ 18	14:15:07	SELECT * FROM ecommerce_data WHERE TotalPrice > 5000 LIMIT 0, 1000	0 row(s) returned	0.015 sec.
✓ 19	14:16:36	SELECT * FROM ecommerce_data ORDER BY TotalPrice DESC LIMIT 10	10 row(s) returned	0.016 sec.

Object Info Session

36°C Partly sunny

Search

Fig 8 : Outlier Detection

Key Insights:

- Successfully performed Exploratory Data Analysis using SQL.
- Identified customer purchasing trends.
- Analyzed payment methods.
- Studied order status distribution.
- Examined referral source performance.
- Evaluated product revenue and sales trends.

Conclusion:

- Successfully completed Exploratory Data Analysis on the E-Commerce dataset.
- SQL queries helped uncover valuable business insights.
- The project strengthened practical skills in SQL, data analysis, and data interpretation.

THANK YOU

